## DEFINITIONS OF CONCEPTS IN EDUCATION: GOALS, LEARNING OUTCOMES, TEACHING AND LEARNING

### *1.2* Lesson Learning Outcomes

By the end of this lesson, you will be able to:

*1.2.1* Define goals, learning outcomes, teaching and learning

*1.2.2* Explain the relationship between teaching and learning.

*1.2.3* Explain the nature and components of teaching and learning

### *1.2.1* The definitions of goals and learning outcomes

Goals from education perspective can be defined broadly as statements of educational intension by a given education system. . They describe skills, attitudes and competencies students should attain during a given education program.

In Kenya, there are eight goals which educational programs from primary to university seek to achieve. They include but are not limited to the following;

1. Education should promote patriotism, nationalism and national unity.
2. Education should promote social, economic, technical and industrial needs for national development.
3. Education should promote individual development and self-fulfillment 4, Education should promote sound moral and religious education.

However, in some countries goals can also be specific. In this case, they are formulated as statements with both an action verb and the competency required after a learning process. In such a case the goal takes the place of an objective e.g by the end of the course the learner/ students should be able to explain/discuss …

Learning outcomes refer to statements of what the learner should know and be able to do to demonstrate their knowledge, understanding, attitudes, values and skills and/or competences at the end of a learning process. Under the Competency Based Curriculum ( CBC)in Kenya they have replaced objectives.

The learning outcomes like goals can be limited to a lesson, course, program or a degree or

level of education.. Therefore, they can be specific and narrow or general and broad. For example learning outcomes for early education according to Nyamoti (2018) include among the following:

*1.* Apply digital literacy for learning and enjoyment

*2.* Demonstrate basic literacy and numeracy skills in learning

3.‟Apply creative and critical thinking skills in problem solving

4. Practice hygiene, nutrition, sanitation, safety and nutrition to promote health and well being.

5, Practice appropriate etiquette for interpersonal relationships

It can also be specific to a lesson e.g By the end of the lesson you will be able to describe

### ***1.2.2*** **Concepts of teaching and learning**

Teaching comes from the old English word „‟tǣcan‟‟‟whichmeanstoshoworpointout. Learningontheotherhandcomesfromtheold Englishwordleornianwhichmeanstoget knowledgeortothinkaboutit.(Brown 2018)

Lefancons (1991) in Nasibi (2003) defines teaching as a process that facilitates changes in learners and entails telling and persuading, showing and demonstrating, guiding and directing learners‟ efforts or a combination of these actions. Learning on the other hand is defined by Ferrant (1980) in NasibI (2003) as a process by which knowledge, ,understanding , attitudes , skills and capabilities can be acquired and retained..

Teaching and learning are therefore interconnected terms. One has to learn before teaching and therefore teaching is dependent on learning.

### *1.2.3* **Relationship between teaching and learning**

The relationship between teaching and learning is crucial. Both of them are processes, are interdependent and inseparable. Teaching and learning as a process starts with planning for teaching, the execution of the teaching and lastly carrying out assessment to find out if the teaching and learning outcomes have been realized.

Teaching and learning are interconnected and interdependent. They cannot operate independently. There can be no learning without teaching nor there teaching in the absence of learning. Both teachers and learners bring in knowledge of their learning experiences and interactions creating understanding and new experiences.

Although teaching determines the nature of learning and effective teaching leads on effective learning , a teacher cannot make learners to learn. The best he/she can do is to create an environment which will lead to learning; an atmosphere that will motivate students and contribute to active learning

According to Brown (2018), in teaching and learning relationship, the teacher has the role of a facilitator, human relation specialist and a motivator. The effective implementation of these roles will promote learning.

### *1.2.4* The nature and components of teaching and learning

The nature of teaching and learning is not only complex, dynamic and flexible but also, interactive and creative. Although a teacher can be seen teaching , learning cannot be seen but can be inferred from change in observable behavior. It is difficult to understand how teachers teach and how their teaching is perceived by their learners. The nature of learning is characterized by the following: universality, experience, continuity, relative permanent change in behaviour and unobservable.

According to Nasibi (2003) the components of teaching and learning include stating the learning outcomes, identifying the content to be taught, selecting appropriate teaching methods and techniques, identifying instructional resources, presenting the lesson, measuring and evaluating the learning outcomes.

## INSTRUCTIONAL MODEL: SYSTEMS APPROACH TO TEACHING AND LEARNING

### Introduction

Systems thinking involves moving from observing events or data, to identifying patterns of behavior overtime, to surfacing the underlying structures that drive those events and patterns. By understanding and changing structures, we can create satisfying, long-term solutions to chronic problems in education. Systems thinking perspective requires curiosity, clarity, compassion, choice, and courage. This approach includes willingness to see a situation more fully, to recognize that we are interrelated, to acknowledge that there are often multiple interventions to a problem in education, and to champion interventions. This lesson explores how to look at education and classroom teaching as a system.

### *2.2* Lesson Learning Outcomes

**By the end of the lesson, you should be able to:**

*2.2.1* Define a variety of terms in systems approach

*2.2.2.* Discuss various components of a system in education

*2.2.3.* Apply systems thinking in instructional planning

### *2.2.1* Definitions: System

Romszowski (1981): “a set of elements or components or elements or objects which are interrelated and work towards an overall objective”

Groenewegen (1993): “a complex of factors interacting according to an overall plan for a common purpose or goal"

"an organized set of doctrines, ideas, or principles usually intended to explain the arrangement or working of a systematic whole" (Merriam-Webster Dictionary)

Generally: Any collection of **interrelated parts** that together constitute a larger **whole.** These component parts, or **elements** of the system are intimately linked with one another, either directly or indirectly, and any **change in one** or more elements may **affect** the overall performance of the system, either beneficially or adversely. The systematic approach to teaching provides a method for the **functional organization and development of instruction.**

**Examples of systems**

Common systems used in our day to day activities are for example:

**The human body** – and subsystems such as the circulatory system, digestive system and nervous system

**Mechanical systems:** refrigeration system, generators, music system, computer system, engines etc.

**Social systems:** organisation such as family, education systems, political parties, trade union movements, industries and churches

**Natural systems:** rivers, forests etc.

General Systems Theory

General systems theory (GST) was outlined by Ludwig von Bertalanffy (1968). Its premise is that complex systems share organizing principles which can be discovered and modeled mathematically. To quote Bertalanffy, ""...there exist models, principles, and laws that apply to generalized systems or their subclasses, irrespective of their particular kind, the nature of their component elements, and the relations or ""forces"" between them.."" (Bertalanffy, 1968, pp 32).

**Systems theory** is the interdisciplinary study of systems. A system is a cohesive conglomeration of interrelated and interdependent parts which can be natural or human- made. Every system is bounded by space and time, influenced by its environment, defined by its structure and purpose, and expressed through its functioning. Changing one part of a system may affect other parts or the whole system. The goals of systems theory are to model a system's dynamics, constraints, conditions, and to elucidate principles (such as purpose, measure, methods, tools) that can be discerned and applied to other systems at every level of nesting, and in a wide range of fields for achieving optimized **equifinality**.

**History of General Systems Theory**

Systems Theory comes from the General Systems Theory proposed by the biologist Ludwig von Bertalanffy (1968).

There is a need for a **unified** and **disciplined inquiry** in understanding and dealing with increasing complexities in organisations.

The systems view investigates the **interaction between components of an organisation, and the relation of components to their larger environment.**

**Characteristics of general systems theory**

- Open system: a system keeps evolving and its properties keep emerging through its interaction with environment
- Holistic view: systems theory focuses on the arrangement of and relations between the parts that connect them into a whole. The mutual interaction of the parts makes the **whole bigger than the parts** themselves.
- Goal-directedness: systems are goal oriented and engage in feedback with the environment in order to meet the goals. Also, every part of the system is interdependent with each other working together toward the goals.
- Self-organizing: productive dynamic systems are self-organizing this implies the adaptive ability of the systems to the changes in the environment.

**Components of a system**

1. Goals and mission: every system has a goal (target to be achieved). All members work towards the achievement of the set goal.
2. **Elements:** there are several elements – these are interrelated and interact. Exists in a hierarchy of relationships – each functional unit forms a subsystem with elements that

cannot work independently. A supra-system: one that has subsystems each of which should perform but is interdependent on other subsystems.

3. Has boundaries – which distinguish it from other systems parts ensuring that a specific task is performed at a particular point, but both/all parts depend on each other

4. Environment: a set of conditions such as resources, changes and constraints (e.g. inadequate material, dissenting voices) that affect the system either positively or negatively. There is a flow both inward and outward the system into the environment. In order to survive, the system must interact with and adjust its environment and the other parts of the supra system. The environment provides inputs such as money, people and resources which will determine the quality of the operation of the system
5. **Harmony:** a coherent interaction for attaining the common goal. Elements work in harmony, although each has its own function which it contributes in achieving the goal of the system.
6. Some degree of disorder (**internal entropy**) – because systems are open and operate within environmental constraints; there is a critical point at which a system may collapse (hence need for flexibility of objectives of a system to minimize entropy)
7. Feedback – input or information about the achievement (output) for purposes of re-examining the system. Provides for assessing the suitability of the success level in the interaction leading to the attainment of the intended goal.
8. Growth- through either transformation or diversification or multiplication
9. Dynamic stability (equilibrium) with the environment – in harmony with the environment and has to exhibit levels of balance among its elements/subsystems
10. It has a set of inputs (learners) which are subject to a process in order to attain certain objectives which appear as outputs (educated students).
11. It has a self-adjusting combination of interacting people/institutions and things to accomplish a pre-determined purpose. (Hooper (1971)

**Impact to Educational Systems: Systems Approach to Teaching/ Instruction** According to Banathy (1987), there are four subsystems in any educational enterprise:

1. The **learning experience subsystem**: the cognitive information processing of the learner
2. The **instructional subsystem**: the production of the environment or opportunities for learners to learn by the instructional designers and teachers
3. The **administrative subsystem**: decision making of resource allocation by the administrators based on the instructional needs and governance input
4. The **governance subsystem**: the production of policies which provide directions and resources for the educational enterprise in order to meet their needs by "owners"
5. The instructional system is part of educational system. Reigeluth (1996)

**Inputs to the educational systems approach include:**

- Well defined objectives,
- Analysis of the intended audience,
- Special criteria desired by the customer (government, parents, students),
- Analysis and use of existing resources - Content requirements, Content development and Use
- A team of instructional system specialists, subject matter experts, writers, and SNE specialists

**The School as a system**

- It is an organised entity with goals received from society (MoE – see national goals, curriculum and syllabuses); receives raw materials (learners) from the environment; learners are put through a process; the output is assessed to see if behaviour is transformed.

- It has elements/parts which must work together for the goals to be realised (head teachers, teachers, support staff, learners, T/L resources, physical resources etc). All these must interact in order to realise the desired output.



- It must have harmony – all elements (all those involved in the education system) must work harmoniously for goals to be realised. E.g. Head teacher as chief administrator must provide leadership etc. Learners must work alongside teachersetc.   Support staff must perform their respective roles etc.

- It must provide feedback – both teachers & learners must get feedback to check if the products are being processed desirably. Feedback maybe positive or negative. Then teacher can adjust teaching methods, resources, experiences, classroom management to achieve desired goals.

**Why Teaching is considered a system**

- Because it has a well-defined goal or objective;
- It consists of more than one element • All these elements work in harmony.
- It also has provision for feedback.

NB: There is an interrelationship among all the components. The focus of the systematic instructional planning is the STUDENT/LEARNER.

**Parts of a systematic instruction explained:**

The following procedure guides the teacher in accomplishing instructional goals:

1. Specification of objectives: Define Objectives - Instruction begins with the definition of Instructional Objectives that consider the learners" needs, interests and readiness.

2. Selection of content: according to the curriculum

3. Assessment of entry behaviour: organizing students into groups.

4. Choice of teaching strategy - On the basis of the objectives, the teacher selects the appropriate teaching methods to be used. And, in turn, based on the teaching method selected, the appropriate learning experiences and appropriate materials, equipment and facilities (resources) will also be selected. The strategic method or learning methodology is based on the nature of the learner. Chose appropriate experiences; allocation of time and learning space.

   - Assign personal roles – The use of learning materials, equipment and facilities necessitates assigning the appropriate personnel to assist the teacher and defining the role of any personnel involved in the preparation, setting and returning of these

1
0 learning
resources.

5. Implement the instruction - Carrying out or Executing teaching instruction. With the instructional objectives in mind, the teacher implements planned instructions with the use of the selective teaching method, learning activities, and learning materials with the help of other personnel whose role has been defined by the teacher.

- o ExamplesofLearningActivities:Reading, Writing, Interviewing, Reporting or doing Presentation,

Discussing, Thinking, Reflecting, Dramatizing, Visualizing, Creating,

Judging, evaluating o ExamplesofLearningResourcesforInstructionalUse:Textbooks,

Workbooks, Programmed materials, Computer, Television programmes, pictures, Slides and

transparencies,

Maps, Charts, Cartoons, Posters, Models, Mock ups, Flannel board materials,

Chalkboard, Reals objects o 6. Evaluate outcomes/Evaluation of teacher and learner performance (quality control monitoring, performance testing etc.) - after instruction, teacher evaluates the outcome of instruction. From the evaluation results, teacher comes to know if the instructional objective was attained.

o 7. Analysis of feedback by teacher and learner: refining the process - It is dependent on the outcome. If the instructional objective was attained, teacher proceeds to the next lesson going through the same cycle once more. If instructional objective(s) was not attained, then teacher diagnoses what was not learned and finds out why it was not learned in order to introduce a remedial measure for improved student performance and attainment of instructional objectives. The teacher should device plans (remedial, tutorial or give more examples) for the learners to fully understand the lesson. (Glaser,

1962) o NB: The purpose of a system instructional design is "to ensure orderly relationshipsand interactions of human, technical, and environmental resources to fulfill the intended goals which have been established for instruction." (Brown, 1969) o The phases or elements are connected to one another. If one element or one phase of the instructional process fails, the outcome, which is learning, is adversely affected. The attainment of the learning objective is dependent on the synergy of all the elements and of all the factors involved in the process

o **Planning to teach as an example of part of the instructional system** o **The lesson plan**

o The teacher/instructor should plan to teach through the scheme of work (long-term plan for content activities and resources in teaching e.g. in one term or year).

- For immediate classroom use, the teacher/instructor prepares a lesson plan. This is a plan indicating the instructional objectives, time required, the major steps to follow, the content and relevant activities, the necessary resources and teaching aids. The plan also gives room for self-evaluation.
- Every lesson plan or lesson presentation must have objectives, elements, harmony (suitable methods, resources, entry behaviour have to be in harmony), evaluation.



**Lesson presentation as a model systems approach**

A well-presented and organised lesson should exhibit the following qualities:

- Relevance to objectives.
- Accuracy of information/content.
- Likelihood to arouse and maintain interest.
    - Learner participation in: Oral activities, Written experiences, Listening activities, Visual activities, Art and craft activities e.t.c.
- Appropriate pace for learners.
- Clarity of organization of information - clarity of expression, conversational, of quality.
- **Models**

□

- Figure1:AsimpleModelofasystemsapproach
- In Figure 2 below, the system consists of four distinct elements A, B, C, D, which are related to or dependent upon each other as indicated. Note that some interrelationships may be twoway, while others may be one-way only. These elements may themselves be capable of further breakdown into other smaller components, and may thus be regarded as **sub-systems** of the overall system.
- Asimplesystemisilustratedschematicalyin Figure2:

□

- **Figure2.Atypicalsystem**
- The processes of teaching and learning can be considered to be very complex systems indeed. The **input** to a given teaching/learning system consists of people, resources and information, and the **output** consists of people whose performance or ideas have (it is to be hoped) improved in some desired way. A schematic representation of systems of this type is shown in Figure 3:

□

**TEACHING STRATEGIES AND THEORIES - HEURISTIC AND EXPOSITORY STRATEGIES IN TEACHER EDUCATION**

## 1.1 Introduction

In this first lesson, we lay the foundation for the entire course by defining the concept entrepreneurship and other related terms. Throughout our teaching experiences, we have found that an understanding of the basic principles behind the subject and their applications increases the students‟ motivation for the subject. We therefore introduce you to the theories upon which entrepreneurship study is based and encourage you to engage in in-depth discussions of these concepts and theories. Since the term entrepreneurship is closely associated with small businesses, we introduce the concept of small business too. The purpose of this lesson is to enable you understand fundamental concepts and theories

of entrepreneurship.

### *3.2* Lesson Learning Outcomes

By the end of this lesson you will be able to:

*3.2.1* Identify Heuristic Methods of teaching

*3.2.2* Identify Expository Methods of teaching

*3.2.3* Apply Kolb‟s Theory of Experiential learning to teaching

*3.2.4* Explain the conveniences and inconveniences of Heuristic Methods

***3.2.5*** **Explain the advantages and disadvantages of 3.2.1 Heuristic Methods of teaching**

**Definition Strategy in teaching**

A strategy in teaching can be defined as the general idea, or the overall way in which the process of teaching and learning is organized. In teaching, one finds some common aspects of practices across a variety of methods. One can therefore say that a strategy encompasses a number of methods which share common features, even though the features may apply differently, in each method.

**Definition of Method in teaching**

A method can be described as a way, which, practical in many instances, a teacher follows in the delivery process of content in a given strand or lesson. It is seen as a means to achieving the learning outcome, in specific steps, deliberately arranged to lead to a desired aim, goal or end.

**Heuristic Strategy**

The heuristic strategy, commonly called "Discovery Strategy" refers to teaching approaches or methods which encourage the teacher to allow learners to discover facts and information, in the process of instruction. In this strategy, the learner and not the teacher, is the center of instruction .The term "heuristic" has been coined from the Greek word <eureka> meaning to find out or to discover. The teacher then becomes a facilitator, who through the skills acquired in his/her training, guides the learners to find out for themselves the answers to issues, rather than telling them directly.

### *3.2.2* Expository Strategy

The Expository Strategy involves direct teaching. The teacher does most of the talking. In the meantime, learners remain largely inactive. In this Strategy learners have to memorize information with the aim of reproducing the same, when called upon to provide explanations. As expected, learning with this strategy will most likely produce a student who has weaknesses in making value judgments on what they have learnt, among others. This strategy leads to shallow learning since it does not lead learners to express their views and opinions on what they are told. The Expository Strategy can be said to be teacher –centered in nature.

### *3.2.3* Experiential Learning

**Definition of Experiential Learning.**

Experiential Learning can be defined simply as <Learning through experience> or <learning through reflection on doing>. Key words here are **reflection** and **doing.** The learner then is not just simply doing something but up and above that he/she is reflecting or thinking, evaluating, judging, at every step of the way in the learning process.

**Kolb's Theory of Experiential Learning**

A number of theories have been developed over the last few decades to support this Learning method. One of the most applied theories applied with this method is **David Kolb's Theory of Experiential Learning.** It is one of the best known educational theories in education. The theory presents a way of structuring a session or a whole course using a learning cycle. The different stages of the cycle are associated with distinct learning styles. Individuals differ in their preferred learning styles, and recognizing this is the first stage to developing a strand that will cater for the different learning styles represented among your learners. **.** In this Lesson we will attempt to see how this theory can be applied to teaching and learning to improve learning .

**The Experiential Learning Cycle**

Kolb's experiential learning style theory is typically represented by a four-stage learning cycle in which the learner comes into contact with all the contents of the programme. These stages are as outlined below.

***1.*** **Concrete Experience** - a new experience or situation is encountered, or a reinterpretation of existing experience.

***2.*** **Reflective Observation of the New Experience** - of particular importance are any inconsistencies between experience and understanding.

**3. Abstract Conceptualization** reflection gives rise to a new idea, or a modification of an existing abstract concept (the person has learned from their experience).

**4. Active Experimentation** - the learner applies their idea(s) to the world around them to see what happens.

## 4.1 INSTRUCTIONAL METHODS: COMPETENCY BASED TEACHING METHODS

In chapter four of this module, we learnt about teaching strategies and teaching methods. We defined heuristic and the expository strategies of teaching and explained that heuristic strategy is a learner centred approach to teaching whereas the expository strategy is a teacher centred approach to teaching. Under each strategy, we discussed several teaching- learning methods which you can use to present information to the students. We concluded that no one teachinglearning methods is suitable for every teaching situation.

In this topic, we shall explore more instructional methods that lean towards competence- based teaching-learning methods. In particular, we shall learn about enhanced teaching methods such as problem solving, innovative, collaborative and cooperative learning methods, projectbased learning and service learning. However, before we delve much in the methods, lets first explore the concept of competence-based teaching methods.

### 4.2 Lesson Learning Outcomes

By the end of this lesson, you will be able to:

4.2.1 Explain competence-based approach to learning and give its benefits.

4.2.2 Identify the enhanced teaching methods and explain how they enhance learning.

4.2.3 Explain each of the six given innovative methods and give their benefits.

4.2.4 For any four of the six given learning enhancement methods give the steps involved in using them.

### *4.2.1* Competence Based Teaching Methods

In our earlier topics, we learnt that a teaching method comprises of the principles and methods used by a teacher to enable the students acquire knowledge and skills. A teaching
-learning method is determined by the complexity of the subject matter, the instructional media available as well as the learning needs of the learners and the objectives to be achieved.

Competency based educational (CBE) approach allows learners to advance at their own pace regardless of the learning environment, based on their ability to master the skills. This approach is tailored to meet learning abilities and can lead to more efficient student outcomes. CBE recognizes that learning can happen in many different ways and rewards demonstration of competency rather than the time spend learning.

### What Is Competence Based Education?

Competence based approach in education refers to teaching and learning of measurable skills rather than abstract learning. The task to be completed must create the outcome that can be identified and measured. The learning task needs to be meaningful and engaging and to be completed in an effective and efficient manner.

Competency based learning has been described as very flexible since it depends on the individual learner. It is skill based and the focus is not the final outcome but the journey. In addition, the student is more engaged in the learning process besides

### *4.2.2* Enhanced Teaching Methods

The following six learning methods are given as good examples of competency-based learning methods.

i Problem solving learning. ii

Innovative learning *iii*

Collaborative learning

*iv* Cooperative learning (case studies) v Project based learning method vi

Service based learning method

Let us now look at each of these learning methods as we explore more on the competence- based learning.

**Problem Solving Learning.**

Problem solving learning is defined as the act of defining a problem, determining the cause of the problem, identifying, prioritizing and selecting alternatives for the solution and implementing the solution. The student is engaged in the entire journey from defining the problem up to the implementing the solution. A four-step problem solving process and methodology is often cited as very effective in this learning method

**Problem Solving Learning Method**

Problem solving approach is a student centred learning strategy which requires the student to become an active participant in the learning process. The student is presented with a problem which requires him/her to find the solutions (searching for information).

The student observes, understands, analyzes, interprets and finds solutions and performs applications that lead to holistic understanding of the concept and in the process, new knowledge is learnt. Through the method, scientific process skills are developed.

**Steps Involved In The Problem-Solving Learning Method.** The

process of problem solving involves five basic steps.

*i* Assessment

*ii* Problem identification, iii Planning iv Implementation and v Evaluation.

**Benefits Of The Problem-Solving Learning Method.**

Problem solving skills are among the most valued skills because they can be applied in various real life situations. Life is about solving problems. Learning about problem solving approaches therefore would equip you with tools to help you strategize solutions when in difficult problem circumstances.

**Project Based Learning Method**

The project method of learning is one which enables students to acquire and apply knowledge and skills to define and solve realistic problem using a process of extended inquiry. In the method the teacher encourages the student to apply their growing skills in purposeful ways. It plays an important role at the initiation, implementation and culmination parts of the project.

**Benefits Of The Project Method**

The primary aim of the project learning method is to make students think independently as they use acquired knowledge and skills to solve problems. As a learning method, the project method has several benefits.

### *4.2.3* Innovative Learning Methods

An innovative learning method is about solving given problems in a different way. This calls for creativity in the problem-solving skills. Innovation therefore involves a different way of looking at a problem and solving it.

**Collaborative Learning Method**

This is a learning method that groups students to work together towards a common goal. The learning method is based on the collaborative learning theory which emphasizes that:

a. knowledge is a social construct.

b. Educational experiences involve interaction and social exchange.

c. Contextually relevant and engaging learning activities are student centred and lead to deeper learning.

In addition, experts claim that:

i the active exchange of ideas within groups of students promotes critical thinking. ii

Teams that engage in cooperative learning achieve;

- Higher levels of thought and
- Retain information longer than students who work solely as individuals.

**Strategies For Effective Collaborative Learning**

Collaborative learning is hard work which does not just happen. It has to be intentionally designed so as to be effective. There are strategies to promote effective collaborative learning.

**Advantages Of Collaborative Learning.**

This method helps students learn how to work effectively as a team. Cooperative learning is a specific kind of collaborative learning where students work together in small groups on a structured activity.

The students are individually accountable for their work and the work of the group as a whole is also assessed. Cooperative groups work face to face and learn to work as a team.

**Service Learning.**

Service learning actively involves students in a wide range of experiences, which benefits others and the community while advancing the goals of a given curriculum.

It offers direct application of theoretical models.

Community based service activity are paired with structured preparation and students reflection. Proponents of academic service learning feel that:

*i* Real world application of classroom knowledge in a community setting allows students to synthesize course materials in more meaningful ways.

*ii* Common goals achieved by service learning are;

- Gaining deeper understanding of the course/curricular content.

- A broader appreciation of the discipline.

- An enhanced sense of civic responsibility.

**Qualities Of Service Learning**

Several qualities of service learning have been given as follows: i

Reflective

*ii* Integrative.

*iii* Contextualized

vi Strength based

v Reciprocal

vi Life long

Some examples of service learning are as follows:

i Field trip

ii Research project

iii Events

iv Crafts

v Fundraisers

vi Awareness campaigns

vii Various service projects such as cleaning, building and gardening

**Benefits Of Service Learning**

The following are some of the benefits of service learning.

i Enables development of skills in critical thinking, problem solving, leadership, decision making collaboration and communication.

ii Building of relationships with community members. iii Connecting experiences to academic subjects. vi Developing deeper understanding of themselves and empathy and respect for others

## *5.1* INNOVATIVE LEARNING STRATEGIES FOR MODERN PEDAGOGY Introduction

The $21^{st}$ century is characterized by great strides in the fields of technology and scientific discoveries, globolization, internetization, informatization and explosion of knowledge and artificial intelligence. In particular, the computers, smart phones, video games, internet search engines have all redefined how we access information. The emerging digital technologies environment has a major impact on how learning in the classroom takes place as visual perception and discussion of knowledge replaces the traditional reading of text books have affected every sphere of life including in education and created immense pressure on how the teacher – learner interaction in the classroom takes place. In addition, the $21^{st}$ century world requires innovativeness and creativity for the individuals and nations to survive in a fast changing global environment. In this global village, Kenya cannot therefore be left holding onto dysfunctional ways which have failed in enabling development in science and technology.

As the students gets exposed to alternative and interactive sources of information, it is not possible for the traditional teaching approaches which heavily rely on promoting rote (memory) learning to remain intact. The teacher has to therefore adapt innovative teaching approaches that are greatly influenced by the changed digital environment

The characteristics of such modern teaching methods are:

i Learner centred

ii Task based or activity based

iii Resource based.

iv Interactive in nature

v Integrative in nature

vi Involve peer collaboration

### *5.2* Lesson Learning Outcomes

By the end of this lesson, you will be able to:

*5.2.1* Explain the meaning of the concepts „innovative learning strategies" and „modern pedagogy".

*5.2.2* Give the characteristics of innovate learning methods and show that each of the eleven provided learning methods as presented under 8.4 to 8.12 qualifies as an innovative learning method.

*5.2.3* Explain each of the eleven innovative learning methods provided and for each give its benefits and challenges to their effective use.

*5.2.4* For the argumentation and the computational learning methods, explain the stages involved and the elements that comprise it respectively.

### *5.2.1* Innovative Learning Methods

An innovative learning method is about solving given problems in a different way. This calls for creativity in the problem solving skills. Innovation therefore involves a different way of looking at a problem and solving it.

### Modern Pedagogy Methods

Modern pedagogy is used to mean learner centred and activity based learning methods, which are used to get learners fully involved. This approach enables the student to construct his/her knowledge and skills through active participation in the teaching-learning process. The teacher acts as a guide, leading learners to achieve the objectives through the activities that the learners are engaged in during the classroom interactions. The following are some learning methods, in which the learner actively participates in the learning process. i Crossover learning ii learning through argumentation iii Incidental learning iv Context based learning. v

Computational learning vi

Learning by doing *vii*

Adaptive learning

*viii* Collaborative learning ix Cooperative learning x

Blended learning

xi E-learning.

We shall now consider each of these innovative learning methods. **Crossover**

**Learning Method**

Crossover learning method is premised on the understanding that learning does not end or stop in the classroom. It involves the teacher proposing a question or problem in the class to be solved in the museum visit or a field trip. The student could tackle the task by taking down notes, collecting pictures, asking other people for their own thoughts, students can learn. The student can then present what he/she has learnt back to the classroom to further illuminate the given problem.

The crossover learning methods has several benefits including enriching textbook knowledge with personal experience.

**Learning Through Argumentation**

Argumentation involves elaboration, reasoning and reflection. Which contribute to deeper conceptual learning. Argumentation is seen as a way of refining information and establishing the truth on an issue or to arrive at a common understanding. The focus is to arrive at a common goal.

students are able to think critically and independently about important issues and contested values by arguing to learn rather than to quarrel.

In the argumentation process, reasoning to justify claims based on evidence or critical evaluation and revision with others is developed. When speakers take different positions, they challenge narratives, tend to make claims and give reasons and evidence to support them resolve the disagreement.

Argumentation is a powerful vehicle for reaching shared understanding. It is very important in research, and in science

**Stages Of Sound Argumentation**

Four stages of argumentation are given as i

Claim

*ii* The data

*iii* The warrant and

*iv* A qualifier

**Incidental Learning**

Incidental learning refers to any learning that is unplanned or unintended. It:

*i* can develop while the student is engaged in a task or activity as a by product.

*ii* can result from reflection on material that was consciously learnt but not recognized as relevant or useful at the time of the study.

*iii* can happen when we least expect e.g. from talking with a friend or playing a video game.

*iv* always happens in the context of another activity or experience

We can therefore see incidental learning as an indirect or additional (unplanned) learning which takes place in the context of an activity or experience.

**Context Based Learning**

Refers to learning whereby real life examples are used in a teaching environment so as to enable learning through actual/practical experience with a subject rather than using mere theoretical information. It is a learner centred approach to learning utilizing scenarios to replicate the social political context of the students or potential working environment.

Context based learning is premised on the understanding that learning takes place when the teacher is able to present information in such away that students can construct meaning based on their own experiences. Learners are therefore enabled to give meaning to concepts, rules and laws by creating connections between the concepts and relating them to their own experiences and the real world. Examples are such learning experiences as internship and service learning.

**Computational Learning**

Computational learning (also known as thinking and problem solving) is solving problems like a computer scientists. The computational learning is the thought process involved in formulating a problem and expressing its solution in such a way that a computer, human or machine can effectively carry it out. It is therefore a problem solving skill whereby, one looks at a problem and solves it systematically and arrives at a solution that both humans and computers can understand.

Computational learning is made up of four elements as follows i

Decomposition

*ii* Pattern recognition

*iii* Abstraction

*iv* Algorithm thinking

### *5.2.2* Learning By Doing Science

Learning by doing is a process of learning whereby students make sense of their experiences. The teachers seek to engage learners in more hands-on creative modes of learning.

The learning method is based on John Dewey‟s (1938) believe that development of practical skills was very crucial to children‟s education. Human beings therefore learn through a hands-on approach.

Asking questions systematically and objectively gathering data and testing hypothesis to find answers is the essence of scientific inquiry. In education, remote labs enables students to understand scientific principles through hands on laboratory experience „learning by doing‟ from wherever they are located.

**Remote Labs**

Remote laboratories (also known as online laboratory or remote workbench) is the use of telecommunication to remotedly conduct real (not virtual) experiment at the:

*i* Real physical location of the operating technology,

*ii* While the student is utilizing technology from a separate geographical location.

Remote laboratories are designed to allow students to remotedly conduct real experiments across the internet. The actual laboratory, materials and the operating equipment are in one geographical location/place while the student is controlling the experiment at a different geographical location sometimes very far apart.

Though remote laboratory applications is relatively new in classroom teaching, Conducting experiments remotely has been in practice for many years. It has the potential to reach many studen**Advantages And Disadvantages Of Remote Labs.**

Remote labs has the following advantages:

i It enhances sharing of knowledge and experiences. ii It reduces startup costs of laboratories iii It improves learning outcomes to support better learning. iv

It increases accessibility to laboratories. v

It decreases fixed and variable expenditure.

However, remote labs have the following disadvantages:

*i* Lack of equipment set up experience.

*ii* lack of hands on trouble shooting experience.

ts who are in remote locations and without laboratory resources.

### *5.2.3* Adaptive Learning

Adaptive learning is a teaching – learning method which uses computer and other interactive teaching devices (algorithms) to orchestrate the interaction with the learner and deliver customized resources and learning activities to address the unique needs of each learner. (Wikipedia). Computers adapt the presentation of educational materials according to the students" learning needs as indicated by their responses to questions, tasks and experiences.

The technology encompases aspects derived from various fields of study including computer science, education, psychology and brain science.

Adaptive learning is driven by the realization that tailored learning cannot be achieved on a large scale using traditional non adaptive approaches.

Adaptive learning endeavors to transform the learner from passive receptor of information to collaborator in the educational process.

**The Importance Of Adaptive Learning.**

Adaptive learning has several benefits some of which are summarized as follows; i

Helps in providing focused attention on an individual. ii

Provides for one on one instruction. iii Creates

time efficiency iv Brings about confidence based

approach v

Creates individualized learning paths. vi Creates personalized learning for

a heterogeneous group. vii

Provides focused remediation.

**Collaborative Learning Methods**

This is a learning method that groups students to work together towards a common goal. The learning method is based on the collaborative learning theory which emphasizes that:

a. knowledge is a social construct.

b. Educational experiences involve interaction and social exchange.

c. Contextually relevant and engaging learning activities are student centred and lead to deeper learning.

In addition, experts claim that:

i the active exchange of ideas within groups of students promotes critical thinking. ii

Teams that engage in cooperative learning achieve;

- Higher levels of thought and
- Retain information longer than students who work solely as individuals.
- **Strategies For Effective Collaborative Learning**
- Collaborative learning is hard work which does not just happen. It has to be

intentionally designed so as to be effective. There are strategies to promote effective collaborative learning.

- **Advantages Of Collaborative Learning.**
- This method helps students learn how to work effectively as a team. Cooperative learning is a specific kind of collaborative learning where students work together in small groups on a structured activity.
- The students are individually accountable for their work and the work of the group as a whole is also assessed. Cooperative groups work face to face and learn to work as a team.

***5.2.4*** **Cooperative Learning**

Cooperative learning is where students of mixed levels of abilities are arranged into groups and rewarded according to the group‟s success rather than the success of an individual member. Sometimes cooperative learning method is thought of simply as group work but groups working together might not be working collaboratively. Cooperative learning is characterized by five elements as follows:

i Face to face interaction ii

positive interdependence iii

individual accountability iv

group processing v

collaborative skills **Types**

**Of Cooperative Learning**

There are three types of cooperative learning. The teacher may use one or more types of groups at a time. The three types are as follows:

i Informal learning groups ii

Formal learning groups iii

Cooperative base groups

**The Benefits And Limitations Of Cooperative Learning**

There is no one best learning method for all learning environments. The Teacher has to be aware of all the available learning methods then select the most suitable or combine several learning methods depending on the concepts being taught, the students learning needs and the instructional media available. The cooperative learning methods has many benefits to the learner but also has some limitations.

**Blended Learning**

This refers to learning which combines technology (online) and traditional learning modes. It therefore combines the strengths of face to face and online learning by extending the reach of the instruction beyond the classroom through the use of digital resources. For example, the instructor may ask students to view a provided video before he/she introduces the concept in class.

**Benefits And Challenges Of Blended Learning**

Blended learning combines the strengths of both face to face and the digital learning. The benefits can be summarized as follows:

i Its both efficient and effective in content presentation ii Its efficient and quick in reaching

a big audience *iii* it saves on travel to attend classes and missing work and therefore it is

cost effective.

*iv* It provides desirable flexibility in presenting the content.

*v* It seamlessly transitions from classroom to computer and vice versa when well used. vi It covers

all learning styles through use of well chosen mediums and techniques. vii It compensates for

limited classroom space.

On the other side, there are several barriers or challenges to blended learning which can be summarized as follows:

*i* expense of technology

*ii* Inadequate training iii Technological issues *iv* The need to adapt content to blended learning v Weakened relationship between the teachers and the students. **E-**

**Learning**

There are many terms used to describe e learning such as distance learning, computerized electronic learning, online learning and internet learning. We can describe e-learning as learning that utilizes electronic technologies to access educational content outside the traditional classroom. It can also be described as the acquisition of knowledge which takes place through electronic technologies and media.

e-learning is interactive (the student can raise his/her hand and interact with the teacher or other students in the class) and therefore use of video tape, TV channel, do not facilitate learning. E learning is conducted on the internet where students can access their learning materials online at an**Challenges To Use Of E-Learning**

The interactive nature of the learning process such as viewing a video, listening to audio and reading through a manual makes the content very interactive and easier to recall information. However, e-learning has its share of challenges including the following:

*i* limited funds

*ii* poor internet access in some areas iii lack of enough trained staff y place and

any time.

**6.1 CHANGING TRENDS IN TEACHER EDUCATION Introduction**

Human beings are dynamic in various spheres of their existence. In education, humankind is constantly looking for ways to improve teaching and learning engagements. Through research and social interaction, various strategies and techniques are often generated to improve teacher education in general and classroom instruction in particular. Various policies and instructional practices are established to ensure that teachers, learners and relevant stakeholders at large are updated on what is in vogue.

In keeping abreast with what is in tandem with modern day goals of education, the educator in general and the classroom lecturer in particular, needs to understand both past practices and future aspirations. This way, appropriate focus will be maintained as the school, college or university seeks to uphold expected performance standards of the learners.

## *6.2* Lesson Learning Outcomes

By the end of this lesson you will be able to:

*6.2.1* Identify key trends in teacher education

*6.2.2* Create interactive learning environments for prospective teachers

6.2.3. Develop and use face-to-face and hybrid digital pedagogies

*6.2.4* Use Open Education Resources in teaching and learning environments

*6.2.5* Plan and implement Large class pedagogic strategies in instruction

**6.2.1 Key trends in Teacher Education**

**Teacher education has for many decades been characterized by centralization of educational programmes and authoritarian delivery of content. In countries throughout the world, the management of education has been under a specific body or ministry that provides the curricula to be taught with minimum learner input. With time however, demands for more inclusive curriculum development and implementation strategies have emerged, paving way to democratic participation in various matters of teacher training and classroom teaching. The current practice of integrating technology in classroom instruction is one such contemporary trend among several others. This should embrace the st**

**21 Century skills6.2.2 Creating interactive learning environments for prospective teachers**

**Traditional teaching and learning (Web 1.0) has largely been characterised by a lot of teacher talk and passive learning. Modern day instruction (Web 2.0) emphasizes learner-centred pedagogy where collaborative, multisensory and proactive learner participation is the norm. Here the use of interactive strategies that encourage constructivism on the part of the student is a key feature of the instructional proc6.2.3 Development of Face-Face and Hybrid digital pedagogies**

In Africa and many other parts of the world, Face –to-Face teaching has been the normal th pedagogical mode of content delivery. However, from the 20 Century, advances in technology have influenced the way teachers implement the curriculum in the classroom. One significant step in using and integrating technology in education is Hybridteaching.

The hybriditymechanism involves the use of **two** dissimilar teaching pedagogies to yield a result that has advantages (and limitations) from both methodologies. In this effort, the delivery of the content is split into onlineand face-to-faceportions. The teacher is supposed to borrow the best ideas from each instructional arena. Hybrid learning is also referred to as Blendedlearning.

*6.2.4* Using Open Education Resources in teaching and learning environments

**Open Education Resources (OER) are a collection of digital technology materials that are publicly available online for educational purposes. They are published under open licenses i.e. anyone can access them and use them as they are or modify them to suit specific needs in education. In teaching and learning, OERs could be in the form of an entire course programme, lecture notes, simulation videos, and student assignments among others. OERs are also known as CreativeCommons. It is of paramount importance for prospective and practicing lecturers to know how to use these resources.**

### *6.2.5* Planning and implementing large class pedagogic strategies in inclusive classrooms

The classroom teacher as the implementer of the curriculum is beset with many challenges.
. One such challenge is on how to undertake meaningful delivery of content to a large

number of students. While it is true that there is no universal definition of the concept of „large classes‟, it is generally agreeable that any class size that a teacher cannot manage effectively and engage the learners actively, may be considered as large. In this regard, inadequacy of teaching and learning resources is a crucial contributor to the level of learning effectiveness as students will not be adequately engaged. Further, a teacher‟s work experience and socio-cultural background will play a major role in how he/she manages large classes.

Despite the differences among teachers on how they create and manage learning environments, it is necessary that all teachers become aware of some of the pedagogic strategies that have been found to work well in various parts of the world including Kenya, for improved learner benefits from the education system.

## SUSTAINING ATTENTION IN THE CLASSROOM

### 1.1 Introduction

In this first lesson, we lay the foundation for the entire course by defining the concept entrepreneurship and other related terms. Throughout our teaching experiences, we have found that an understanding of the basic principles behind the subject and their applications increases the students‟ motivation for the subject. We therefore introduce you to the theories upon which entrepreneurship study is based and encourage you to engage in in-depth discussions of these concepts and theories. Since the term entrepreneurship is closely associated with small businesses, we introduce the concept of small business too. The purpose of this lesson is to enable you understand fundamental concepts and theories of entrepreneurship.

### *7.2* Lesson Learning Outcomes

By the end of this lesson, you will be able to :

*7.2.1* Explain the concept of motivation.

*7.2.2* Explain the types of motivation

*7.2.3* Explain the importance of motivation in teaching and learning.

*7.2.4* State the theories of motivation

*7.2.5* Explain Maslow theory of motivation and its relevance to education

***7.2.6*** Explain ways in which you can motivate and sustain motivation in the classroom**.**

***7.2.1* The concept of motivation** Motivation is essential for effective teaching and learning.

As a concept it can be defined as the reason why people behave the way they do or do

whatever they do. According to

Woodfolks 91992) in Nasibi(2003) motivation is the general process by which behaviour is initiated and directed towards a goal.

There are two types of motivation which explain learner"s behaviour in the classroom: intrinsic and extrinsic motivation. Intrinsic motivation is influenced by the reinforcers that are present in the activity being performed such as joy and satisfaction to the performer. Extrinsic motivation is influenced by outside rewards to the action being performed.

***7.2.2* Importance of motivation in teaching and learning**

Educators have argued that motivation is part and parcel of learning because ;

1. It stimulates and makes learning interesting and sustainable .

2. It increases concentration and alertness among the learners.

3. It improves learners performance both in class tasks and in examination.

4. It improves creativity and initiative among the learners**.**

***7.2.3* Theories of motivation**

Motivation theories though many and varied can be categorized in to behaviourist, humanist, cognitive and social learning theories. The advocate of behavioural was B.F Skinner (1938 ) argued that all behaviours are motivated by rewards. In a learning situation therefore a learner has to be rewarded for making a positive response. This will propel him/ her to move to the next level.

Among the proponents of humanistic theory are Ibrahim Maslow(1943)\and his well known theory of hierarchy of needs. Perumal (2009) posits that in this theory " the single, holistic principle that binds together the multiplicity of human motives is the tendency for a new and higher need to emerge as the lower need fulfils itself by being sufficiently gratified."

Herzberg‟s motivational theory argues that if there is improved learning environment, communication, administration procedures and resources then there will be enhanced positive motivation among learners.

**Expectancy theory is differentiated under cognitive theory and motivation is enhanced when there is a likely hood of success in the activity or task at hand, Related to this is expectancyvalued theory which states that motivation is influenced by the reason for being involved in the tas7.2.4 Motivation techniques in the classroom**

Motivation in the classroom is influenced by the learning environment, the teacher‟s experience and expertise in motivation techniques and the learners‟ nature of motivation whether they are intrinsically or extrinsically motivated.

A conducive learning environment which is free of destructors, neat, pleasant, comfortable, bright, with attractive and relevant displays is a plus for learners motivation ( Nasibi 2003).

The techniques to be put in place include reinforcement techniques, stimulus variation, set induction and introducing a variety, novelty and familiarity in the lessons.

It is important that the teacher maintains success expectation, gives learners opportunity for active response, uses interesting methods, caters for individual differences among the learners. Besides, the educator has to use students present motives, minimize performance anxiety and give immediate feedback.

### 7.2.5 Sustaining motivation

Once motivation has been achieved, the instructor has to ensure it is sustained throughout the lesson. Psychologists argue that the students‟ attention span is short lasting between 10 to 15 minutes on the average. For learning to go on, the teacher has to come up with strategies of maintaining the motivation.

Below are some of the techniques one can put in place to ensure learning goes on:

1. Use of set induction at transition or evaluation stages of the lesson. This can be achieved by use of paradoxical statements, presentation of novel stimuli, use of behavioural surprise etc

2. Applying stimulus variation through focusing ( use of verbal statements, specific gestures, introduction of props, emphatic movements )

3. Utilisation of reinforcement techniques where desirable behaviour is reinforced and undesirable is discouraged through punishment.

4. Use of examples which are simple, relevant, interesting and dramatic.,

5. Establish incentive contingencies .

6. Provide immediate feedback.

### *8.1* EFFECTIVE CLASSROOM COMMUNICATION Introduction

Communication is a unique human activity that we are all involved in. They say communication is the life-blood of any group of people or organization including the family members, church, schools ,communities and nations. Human beings live socially because of their ability to communicate. Lack of communication among people is like the HIV virus or Corona (Covid19) disease. Language is the principal tool of communication but we are to communicate without using language. We may communicate using gestures, symbols, visual forms, written messages, electronic devices, body movements or even manner of dressing as well as remaining silent or mute. Teachers and all other professionals must communicate effectively in order to achieve their teaching and learning goals. This is what is probably known as interpersonal communication .It takes place between the teachers and learners as well as among the learners themselves. The communication process is vital in the classroom teaching and learning process in the classroom. Effective communication will lead to effective teaching which will in turn lead to effective learning.

### *8.2* Lesson Learning Outcomes

At the end of the lesson or lessons the learners will be able to:

*8.2.1* Explain the communications process using the SMCR model.

*8.2.2* Explain the main qualities of a teacher in classroom control and management.

*8.2.3* Identify main barriers to communication

*8.2.4* Apply questioning and feedback methods in classroom communications.

### *8.2.1* Definition of Communication

There are many and varied definitions of communication .Historians, sociologists ,engineers, technologists and other professionals use the term communications differently depending on their contexts and aspects referred. In our teaching and learning context we shall confine the term communication to passing information or ideas from the teacher to the learners as well as receiving the feedback from the learners in the classroom. There are three aspects to be considered in the communication process. These include:

*i.* That, communication involves **people** .This implies that communication tries to understand how people relate to each other.

*ii.* That, communication involves **shared meaning** implying that for effective communication there must be agreement in words and terms used in the communication .This is also called **mutuality** of meanings.

*iii.* That communication may be symbols gestures sound letters numbers and words that represent or approximate ideas that are meant to be communicated.

**Forms or Types of Communication**

In transmitting information or messages from one person to another the following forms or types of communication could be used:

*i.* Written form

*ii.* Oral (verbal) form.

*iii.* Pictorial (visual) form.

*iv.* Gestures, body movements including hands, face, mouth, eye, feet or body posture.

*v.* Manner of dress.

*vi.* Silence or mute.

**Process of Communication (Concepts in Communication)**

The available literature highlights the process of communication consisting or comprising

1. Sender
2. Message
3. Receiver.

The figure below illustrates the three components or parts of the communication process.

## A MODEL OF COMMUNICATION PROCESS

For effective communication the Sender initiates the message to be transmitted .The Sender **encodes** the information with the conviction that there is mutuality or shared meanings between the Sender and the Receiver. In turn, the receiver decodes the information or message and responds to the sender. Between the Sender ( source) and the receiver there can be other modes of transmitting the messages. This is called the **channel** which may send the message in various forms e.g. Verbal and Non-verbal .The receiver should give the **feedback** to the sender for effective communication to take place. Schematically this is referred to as SMCR model where S stands for (source) M for (message) C for (channel) and R for (Receiver).Below is the SMCR communication process:

Source: Twoli et al 2007:105

In the classroom the teacher initiates the communication process the teacher is the source. The learner is the receiver of the message or information. The teacher asks questions to the learners. The students answer the questions. The channel may be verbal or Non-verbal. The answers the questions are the feedback in the communication process.

### Concepts in Verbal and Non-verbal Communication

Verbal communication is the most commonly used channel by teachers in the classroom. For effective verbal communication the following aspects should be considered:

*i.* The teacher should use clear and audible voice to be heard in the whole classroom .The teacher should not shout or speak in too low or soft voice. The voice should be appropriate so that student‟s listen comfortably to be able to be assimilate the information given by the teacher.

*ii.* The pitch of the teachers‟ voice should be appropriate or moderate .The teacher should vary the pitch or tone according to the needs of the classroom and the learners.

*iii.* The voice clarity is also important in verbal communication pronunciation problems may negatively affect the communication of the verbal communication of the teacher.

*iv.* The teacher should face the learners for effective verbal communication to take place.

B. Non-verbal communication is also common in classroom interaction with the learners. These include the use of gestures, facial expressions, eye contact, use of hands and body movements“ .These may send special messages from the teacher to the learners. Both the teacher and the learners may successfully utilize non-verbal communication in the classroom teaching and learning process.

C. To improve and compliment classroom communication the teacher should plan and utilize the writing board blackboard, whiteboard and / smart board. These devices should be used to reinforce and clarify oral communication .The following steps are important in improving teaching and learning:

*i.* Plan the teaching and use of boards in advance

*ii.* Ensure that what is written on the boards is neat, tidy and legible by all learners in the classroom.

*iii.* Write clear and large words phrases and sentences on the board.

*iv.* Clear the boards after use do not write all over the board and avoid confusing the learners.

### *8.2.2* Barriers to communication

There are various conditions that restrict or hinder effective communication in the classroom. These are:

*i.* **Physiological barriers**. These are factors that affect our bodies, both for teachers and learners. They include sickness, deafness, blindness or any impairment.

*ii.* **Psychological barriers**: These include attitudes, interests and expectations of either teachers or learners.

*iii.* **Physical (environmental) barriers**: Noise and other natural conditions may hinder effective communication. Even the physical distance between the sender and the receiver may negatively affect effective communicating in the classroom.

Others barriers to effective communication include the following;

i. Language differences
ii. Lack of feedback *iii.* Lack of trust

*iv.* Social distance or status

*v.* Complexity of information

*vi.* Quantity of information

*vii.* Fear and anxiety

*viii.* Lack of common experience.

*ix.* Gestures

*x.* Prosodic features e.g. stress and intonation

*xi.* Differing perceptions

*xii.* Emotionality e.g. anger, love, defensiveness, hate, fear, jealousy, embarrassment

*xiii.* Noise any factor that disturbs, confuses or interferes with the communication.

*xiv.* Technical jargon – Too many assumptions too long chains of command, wide span of control, inaccurate judgment or failure to communicate at all.

**Effective Communication**

This will be promoted and enhanced by:

*i.* **Trust**- in classroom trust by teachers and learners will lead to effective communication and effective management of the classroom.

*ii.* **Active listening**: Good and active listening demands “paying attention” to teachers and learners before responding on giving feedback.

*iii.* F**eedback** – Without feedback communication it is incomplete and not effective

*iv.* **Effective communication**- demands to have the right messages getting to the right people at the right time.

*v.* There are desirable **channels** of communications either upwards, downwards and sideways. These channels should be used for effective communication ever in the classroom control and management.

*vi.* Effective communication will lead to effective management and control of the classroom which will in turn lead to effective accomplishments of the goals and objectives of the lesson. This the **means** and **end** to effective communication in all the classrooms for both teachers and learners.

### *9.1* SYLLABUS INTERPRETATION AND PROFESSIONAL TOOLS Introduction

One of the most central actions in teaching and learning, particularly in a classroom setting, is the process of preparation. In order for the delivery of content to be successful and seamless throughout the year, term or even the 40 minutes of a strand, the facilitator needs to have clear plans of each of the documents. The most common of these documents are the Syllabus, the Scheme of Work, the Strand plan and the record of work.

### *9.2* Lesson Learning Outcomes

By the end of this lesson you will be able to-:

*9.2.1* Define a school syllabus

*9.2.2* Explain a Scheme of work

*9.2.3* Explain a strand plan

### *9.2.1* The Syllabus

A definition of the Syllabus may be stated as follows-:

A syllabus is a document that shows the content to be covered as well the learning outcomes to be achieved in a given subject within a given period and at a particular level. In Kenya the syllabus is prepared by the KICD (Kenya Institute of Curriculum Development). In order for the Syllabus to be relevant, it has to take into consideration the National Goals of Education, the Learning outcomes of both the subject and the lessons and sub lessons of the subject. **N.BLessonhereisthetopic,accordingtotheCompetencybasedcurriculum (CBC),whilethelessonisnowcaledtheStrand.**

### *9.2.2* The Schemes of Work

A scheme of work is a document that a teacher develops from the curriculum design. A scheme of work shows how the planned curriculum content shall be distributed within the time allocated for the learning area.

A scheme of work helps the teacher to do the following:

-To plan on what resources will be required.

-To decide on the methodology to be used.

-To plan for assessment.

A scheme of work has several components which include the following; week, lesson, strand/theme, sub-strand/sub-theme, specific learning outcomes, suggested learning experiences, suggested learning resources, suggested assessment methods and remarks.

A sample Scheme of Work

| School | Grade | Learning Area | Term | Year |
|---|---|---|---|---|
| ShupavuHigh | Form2 | Thesub-strand | 2 | 2022 |

| Week | Lesson | Strand | Sub strand | Specific learning outcome | Key inquiry question | Learning experiences | Learning resources | Asses<br><br>ment | Reflectio<br><br>n |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | | | | | | | | |
| | 2 | | | | | | | | |
| | 3 | | | | | | | | |

Adapted from KICD

### ***9.2.3*** **The Strand Plan (Lesson Plan )**

A lesson plan is an essential document for effective teaching and learning. A well-done lesson plan helps the teacher to;

-organize the content to be taught in advance focusing clearly on the content to be covered and the way it should be taught hence avoiding vagueness and irrelevance

- plan, prepare and assemble teaching/learning resources

- Present concepts and skills in a systematic manner using appropriate strategies to achieve the stated lesson outcomes

- Manage time well during the lesson

-select and design appropriate assessment methods to evaluate the teaching and learning process

**Organization of learning**

**-** This shows where learning will be taking place. It could be in the classroom, or outside the classroom or a visit to a nearby library or farm.

**Introduction**

**-** The lesson should be introduced in an interesting and stimulating manner to arouse curiosity in the learners**.** Integrate the learners‟ related past experiences as much

**Components of the lesson plan**

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**Organization of learning -** This shows where learning will be taking place. It could be in the classroom, or outside the classroom or a visit to a nearby library or farm.

**Introduction -** The lesson should be introduced in an interesting and stimulating manner to arouse curiosity in the learners**.** Integrate the learners‟ related past experiences as much as possible, tapping into learners‟ prior knowledge to prepare them for additional content you‟re about to introduce.

**Lesson development -** This is the actual teaching of the Learning area content. The subject matter is divided into steps. Each step should contain one main idea or experience.
Explicitly outline how you will present the lesson's concepts to the learners and the activities to be undertaken in each step-in order to achieve the stated outcomes.

It should indicate clearly what and how is to be taught and the learners activities (learning experiences). The teacher should vary the teaching/ learning activities as the need arises.

**Sample Strand (Lesson plan)(AdaptedfromKICDtrainingmanual)**

| School | Learning Area | Class | Date | Time | Roll |
|---|---|---|---|---|---|
| Shupavu High School | | Form 2 | 7/3/2020 | 9:40-10:20 | 40 |

**Strand:**

**Sub-Strand:**

**Specific Learning Outcomes:**

***1.*** **By the end of the sub-strand, the learner will be able to:**

**a)………………………………………………..**

**b)…………………………………………………..**

**c) ................2. Learning resources.**
(listherealresourcesnecessaryforaccomplishment ofthespecificlearningoutcomes.

***3.*** **Co-competencies.** (ListherealtheassociatedvaluesandPCIsthataccompany achievementofthestatedlearningoutcomes)

***4.*** **Organization of Learning**

***i)*** **Introduction:**Introducelessonwithanappropriateset-inductionthatwilldrawthe atentionoflearnerstothestrand

***ii)*** **Lesson(strand )Development**

**Step 1-**Learners react to the set induction

**Step 2-**Teacher uses learners responses to synthesize the sub-strand using key inquiry questions

**Step 3-**Teacher delivers the content of the lesson (use demonstrationandotherelevant methods)

**Step 4-**It is now the learner‟s turn to apply content and prove to themselves, with teacher guidance that the leaning outcomes have been attained**.**

**Conclusion**

The teacher now gives a relevant activity/assignment to learners**.**

**…………………………………..**

Reflection. **The teacher prepares a self-evaluation on what was achieved/not achieved for the purpose of continuity**. **9.2.4 Record of work**

A Record of Work is a document kept by the teacher showing the work that has been done at the end of every lesson, strand or sub-strand. The entries are made daily by the individual teacher. It helps in:

- accountability and transparency of work covered by the teacher

-the continuity of teaching of a class

- that a new teacher has an idea of where to start teaching a class

- the evaluation of schemes of work after a period

- providing uniformity of content coverage in case of several streams

The record tracks the achievement of learning outcomes and the competencies acquired by the learner. The record can be used to show the teacher whether their teaching has been effective in addressing the learning needs of individual learners. It therefore acts as a guide for the teacher to be able to give the required attention to individual learners to ensure the desired outcomes as stated in the curriculum designs are portrayed by all the learners. The progress record can also be used to give the learner and the parents/guardians information about the learner‟s progress.

**Components of a Record of Work**

**Time Frame:** There should be an indication of the date and week when the work was covered.

**Work done:** strand and sub-strand as derived from the specific learning outcome(s)

**Reflection:** The remarks column should have statement(s) specifying the success and or challenges of that lesson and recommendations.

**Details of The Teacher:** include the name, signature or initials of the implementing teacher for accountability.

**INCLUSIVE DIFFERENCES EDUCATION – PROVIDING FOR INDIVIDUAL (Supporting ALL Learners)**

### *1.1* Introduction

In this first lesson, we lay the foundation for the entire course by defining the concept entrepreneurship and other related terms. Throughout our teaching experiences, we have found that an understanding of the basic principles behind the subject and their applications increases the students‟ motivation for the subject. We therefore introduce you to the theories upon which entrepreneurship study is based and encourage you to engage in in-depth discussions of these concepts and theories. Since the term entrepreneurship is closely associated with small businesses, we introduce the concept of small business too. The purpose of this lesson is to enable you understand fundamental concepts and theories of entrepreneurship.

### *10.2* Lesson Learning Outcomes

By the end of the lesson you will be able to

*10.2.1* define the term „Inclusive Education‟

*10.2.2* highlight the main features of Inclusive Education

*10.2.3* list methods of identifying leaners‟ individual differences

10.2.4 give ways of providing for individual differences in your class.

### *10.2.1* The concept of Inclusive Education

The term Inclusive Education refers to education which accommodates learners with various individual differences. The differences could be in terms of their age, gender, and mental abilities. It is important to be aware of the individual differences which exist among your learners, in order to address them during instruction.

### *10.2.2* Types of individual differences among learners

There are various types of individual differences which exist among learners in each given class. They include age, gender, mental ability, physical characteristics, physical challenges, emotional stability and economic status. These differences can make learners learn at different rates. Hence the need to identify learners‟ individual differences in order to reach them at the point of academic needs.

### *10.2.3* Methods of identifying learners with individual differences

This section deals with methods of identifying learners‟ individual differences or special needs. It is important to know your students well in order to identify those who have special learning needs and to cater to the said needs adequately. You can use some or all the following methods to find out the individual differences which exist among your learners.

a) Observation of learners‟ behaviour in class and outdoor activities.

b) Use of oral and written interviews.

c) Use of records, e.g. admission records, medical records, progress records, punishment book, etc.

d) Use of teacher-learners‟ conferences.

e) Use of open days.

**Use of tests to g10.2.4 Methods of providing for individual differences**

You should note that methods of providing for individual differences will depend on the nature of the differences and time available in a given lesson. They include:

a) Use of a variety of instructional resources

b) Use of motivation/encouragement

c) Use of group work

d) Use of individualized instruction

e) Use of remedial work

auge learners‟ understanding of the content covered

**COMPETENCY BASED ASSESSMENT AND EVALUATION**

**1.1 Introduction**

In this first lesson, we lay the foundation for the entire course by defining the concept entrepreneurship and other related terms. Throughout our teaching experiences, we have found that an understanding of the basic principles behind the subject and their applications increases the students‟ motivation for the subject. We therefore introduce you to the theories upon which entrepreneurship study is based and encourage you to engage in in-depth discussions of these concepts and theories. Since the term entrepreneurship is closely associated with small businesses, we introduce the concept of small business too. The purpose of this lesson is to enable you understand fundamental concepts and theories of entrepreneurship.

***11.2* Lesson Learning Outcomes**

By the end of the lesson you will be able to:

11.2.1 Distinguish between assessment, measurement and evaluation

11.2.2. Distinguish between formative and summative assessment and evaluation

11.2.3 Explain the meaning of competency based assessment

11.2.4 Explain the meaning of competency based evaluation

***11.2.1* Concept of assessment** The term "Assessment" has been used interchangeably with

"measurement" and even with

"evaluation". To minimize confusion, assessment has been used in this lesson to refer to various methods and tools used to determine the knowledge and skills, together with the attendant attitudes, learners are acquiring in the course of instruction.

Among the major purposes of assessment include the provision of performance feedback to students, teachers, school administration, parents /guardians and sponsors.
Consequently, appropriate steps are taken to facilitate learner achievement at school.

### *11.2.2* The concept of measurement

Measurement is a common activity frequently employed by teachers in the course of curriculum implementation. After covering a given topic or set of topics a classroom teacher will usually seek to establish what students have understood and how well they have done so. To do this he/she will use tools called tests. Examples of teacher-made test are: Essay, Multiple choice, True/False, Completion, Structured tests and Short answer tests.

***11.2.3*** Concept of Evaluation

You have learned that one can measure the achievement of learning outcomes through classroom tests. This exercise is called **assessment** and utilizes **measurement** to obtain data in the form of scores. In assessment, learners‟ competencies or abilities are determined. Test scores can be used, in combination with other data, to make decisions on specific aspects of interest in education. The making of a value judgment on the worth, quality, effectiveness or significance of an educational programme, course or Unit is referred to as **evaluation**. In order to pass a comprehensive judgment on the quality and relevance of an educational programme or Unit against stated learning outcomes, it is necessary to obtain information from a variety of stakeholders such as learners, teachers, school administrators, parents/guardians, local leaders and employers .among others. A variety of techniques such as questionnaires, classroom observation and interviews are utilized for such a purpose.

### *11.2.4* What is formative assessment?

Formative assessment concerns itself with determining the progress and challenges being experienced by both the learners and their respective teachers/lecturers in the course of covering designated content. By analyzing the student performance through various tests, interviews and observations of student behavior, the interested parties can be able to make the necessary changes and modifications in the subject matter, resources and instructional procedures.

### *11.2.5* Meaning of summative assessment

At the end of each education cycle, such as end of Primacy, Secondary school programme or end of a college/ university semester, it is common practice to determine learner performance. This is usually done through national and college/university examinations.

Such an activity seeks to find out the kinds of knowledge and skills, as well as the attendant attitudes that learners will have acquired. Such an exercise falls in the realm of **summative assessment.**

### *11.2.6* Meaning of Competency based Assessment (CBA)

When one is hired to do a particular task or job, the quality of his/her output or performance will indicate his/her ability to perform the given task. Further, the time taken to complete the assignment will be a measure of competency. In Education too, our competencies are usually judged via various assigned practical tasks or activities; for the classroom teacher/lecturer, this is done through specific assessment engagements, mainly quizzes, examinations or projects. Through these activities, the **skills or competencies** acquired by each learner can be identified both during the process of curriculum implementation and upon completion of a given programme of study. The former stage is referred to as competencybasedassessment(CBA)while the latter is in the realm of competency based evaluation(CBE).

### *11.2.7* Concept of Competency based Evaluation (CBE)

As noted earlier, Evaluation is an activity that seeks to determine the value of an educational programme or process. Therefore, CBE is an undertaking that focuses on the kinds of abilities or competencies that learners have acquired over a definite period of schooling such as at the end of 4 years at secondary school or university. The judgement about learner competencies will be made against specific criteria such as the learning outcomes stipulated for the course content in question.